

Interactive Spatialized Animations in “The Wedding Chamber Project” as a Methodology to Produce Phenomenological Diegetic Renderings Inside an XR Immersive Environment

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ABSTRACT

Animation inside an XR immersive world nowadays implies a strong assertion of what a relatively old media transfigures within a new one. The Extended Reality computing environment becomes the emerging framework and catalyst where animated narratives can construct a spatialized layer of meaningful diegetic experiences revamping on a strong basis what interactivity means for users. In this short paper, these insights are exposed, analyzed, and discussed from the computing, artistic, and interdisciplinary perspectives, referring to “The Wedding Chamber”, an XR project in progress that articulates an experimental way of playing animation to produce knowledge through an interactive methodology

Keywords: Immersive, Usability, HUD, Fourth Wall.

Index Terms: Diegesis— Volumetric Presence —Extended Reality- Immersive Narrative – Interactive Spatialized Animations

1 INTRODUCTION

An XR world not only can spatialize the immersive paradigm beyond a conventional VR/AR permutation, but it basically offers the possibility of restructuring what the relations of virtual space mean in the current context of immersive environments.

It is certain that without a compulsory implementation of a usability standard an immersive prototype would become untenable and very confusing to handle. Therefore, animation will become the spatialized layer of diegetic renderings to integrate what the immersive narrative requires from the user to transition from immersive VR space to the Extended Reality immersive surroundings. This mechanics is preserving the feeling of “plunging” not matter how disruptive the concrete encompassing of the real-world objects would entail.

The *Wedding Chamber* project main purpose was to explore theses dimensions in terms of possible outcomes and insights in the following notions, the conversation between the installation animated diegetic renderings devices aiming user’s awareness of virtual spatial relations, the immersive narrative induced by intelligent programming utilizing navigation meshes to enable the user’s camera being treated as an active “agent”; and finally the implementation of an aesthetic notion of perception to actualize the meaning of lived actions as a “suspension of judgement”, while interactivity is being performed.

2 DIEGETIC AND SPATIAL RELATIONS WITHIN THE XR ENVIRONMENT

The *Wedding Chamber* project development process starts in 2016 after being funded and supported by Indiana University and Purdue University. The project study the aesthetic and proto-technological insights painted by Andrea Mantegna in the Saint Giorgio Castle, in Mantua, Italy, in the 15th century. The subject of study offered a rich range of visual, illusionistic, and narrative implementations

that worked as one of the pioneering proto-immersive environments in western art [1]. The installation development based on this chief masterpiece pursued two associated objectives, the deconstruction of the artwork based on a rigorous interpretation of its imagery, and the implementation of animated programmed devices based on the fresco illusionistic imaging to enable an interactive dialog with the user.



Figure 1. The Wedding Chamber Installation

The project physical structure consists in, and installation based on three panels that work as a mediated layer producing renderings in the headset as the interface of navigation. See Figure 1. However, the project implements the transition from the VR immersive modality to the Extended Reality one, beyond these three virtual and physical panels. The XR part of the installation is a mixed reality area containing holographic animated scenes rendering specific visual stories from the original fresco [2]. This module main goal was to enable through animated characters a network of diegetic renderings, communicating to the user what he perceives from the mapped environment, and how this dialog impacts what the animated characters are unfolding within the story.

A very evident fact is appreciated when users engage in an immersive adventure, what is happening inside the device’s modality of visualization (VR headset), is eventually disrupted by the concrete real-world surroundings to break this experience.

This phenomenon turns the immersion in a somehow ephemeral event, precluding the consummation of what the content has articulated in terms of immersive content absorption through diegesis i.e., the interior vision and experiences of a world, and how this is revealed through the objects and characters narrative [3]. Normally, a standard design modality in VR/AR prototypes provides Head Up Displays to induce and guide experiences, however, this concept does not differ greatly from what a conventional interface entails. Then, it must seek the efficient articulated transition from the VR to the XR immersive world navigational paths, besides that, it must establish an interactive dialog with the user providing meaningful visual insights to produce a continuing engagement despite the disarray impact from real world objects and events. These initial findings in the installation usage, lead eventually to the incorporation of the “Fourth Wall” i.e., the metaphorical barrier that separates the

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user/performer and the fictional world, as a significant event to trigger the user's awareness through a layer of animated virtual spatial relations required to accomplish the immersive content continuity within the experience.

2.1 2D and 3D animation becomes part of the fourth wall

How 2D and 3D animations would become critical components of the concept of the "Fourth Wall" deemed as a conceptual reference? The disrupting approximation of the "Fourth Wall" concept was used in the Truffaut's *Fahrenheit 451* movie to represent interactive screen devices inside a home room, that engage the female protagonist in a virtual relationship with the characters and situations happening inside of the so called in the movie "Wall Parlour" system. The novel element in this fictional technological forebear implied that the virtual content as a source of immediate stimuli, turned the viewer in the emitter source of responses to those events. The Fourth Wall metaphor alternation becomes meaningful and effective when the content in the virtual world appeals the users' awareness [4]. Unlike gaming interaction, the virtual content is producing a conversation where the usability's layer of engagement develops an intermutually required communication bond.

In the installation project, the usability outcomes during its testing process, gave several indications that users' primeval perception is at first disorientation. This is a normal initial behavior, however, the layer of animated diegetic renderings based on 2D/3D animations provided the conceptual and the physical map helping to guide the user and making the navigation a concurrence action.

The integration of 2D and 3D animations becomes a versatile utility for a more effective imaging to replace a likely overcrowded head up display. However, it is necessary to distinguish the intention of implementing a layer of diegetic renderings instead of the embedding of a classic HUD interface. The goal for the implementation of a dynamic layer of diegetic rendered animations was primary the notion of integrating the computing phenomena with its variables, and the narrative context of the installation.

Every element of the constructed environment inside the VR immersion, and the extended mixed/overlapping world has become an active layer of interaction and conversation with the user. In the end the user experiments the concept of full immersion whether this occur in the VR modality, or in the Extended one, being this presence achieved by a consciousness of what happens in the virtual world like something that involves a full judgement of events.

3D animations display a volumetric presence of objects through the integration of haptics i.e., the tactile presence/perception of objects in computer simulations. Then it is the awareness of the scene performance through the user's grasp that eventually live the world, this becomes the full immersion. The fourth wall that separates the user's presence is permeated when the diegetic communication occurs. The Fig. 2 shows how animated interfaces engage the user to approach the narrative in the installation.

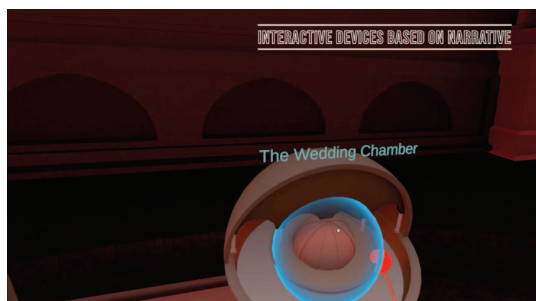


Figure 2. 3D Animated diegetic devices

In the installation architecture, the objects that possess an encapsulated programming component to produce responses from the users, are intended to capture the camera vector orientation toward the object is broadcasting a symbolic rendering. This programming component is implemented by the transform values of *Vector 3 XYZ*. The engaging dynamic develops a tracking behavior as well if the users are approaching a responsive object. Then the object starts the dialog and begin following the user's camera that attracts the object vector orientation through transform detection and *Vector 3 Lerp* i.e., linear gradual interpolation between to vector points. The "object following" behavior acting in the installation, has embedded a mechanical animated action, also the characters from the frescos' stories run the same programing, gazing toward the user's viewpoint of the VR/ XR camera rig

2.2 3D animations that trigger behaviors via camera rig collisions

The primary navigational behavior in the VR/XR integrated world works together with the camera rig collision. This dynamic is determined by the approachable object that contain colliders as well.

In the VR environment objects that operate inside this layer of diegetic renderings, become in fact a network of visual messages emitted from the objects containing colliders. Whenever the camera rig approaches an object with diegetic content, the content behavior activates. In simple terms this is called detection, and the more interesting feature of this action, is the configuration of this exposure as the presence of the camera rig. When the transition to the XR world occurs, an additional component of collision is determined by the mapped environment. This is a very strong component to produce awareness about how the holographic 3d animated content is fostered by an eventual physical component of the mapped environment. Unlike the VR world, where the play area established by the "chaperon" boundaries fades as the player leaves, in the XR world therefore, the passing through is activated, then the transition to the XR immersion happens.

To produce this action, the Meta (oculus device) headset (and eventually the HoloLens device, implement the Mixed Reality functionality), then the device start mapping the environment. It is very important to acknowledge that whenever the whole installation is transported to a different location, the application begins a new mapping, therefore the holographic objects are relocated according with the new mapped environment. Both, the VR and the XR camera rigs, can interact with the environmental colliders, then the holograms are standing and responding. See Figure 3.

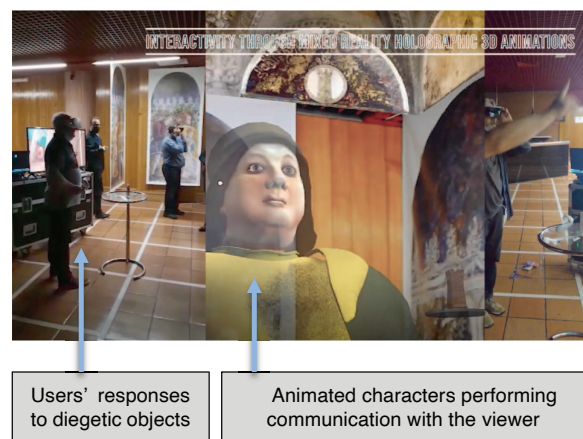


Figure 3. 3D Animated diegetic devices

The visualization technology provided by the Microsoft HoloLens, enable a proprietary system to configure and implement the play area by three dimensionally mapping the environment. Therefore, the design of the immersive/mixed reality environment is developed by the MRTK (Mixed Reality Tool Kit) platform. Meta Quest 2 (formerly Oculus) can perform the passing through from the VR world to the mixed Extended Reality modality as well.

A very interesting and uncanny phenomenon is the one that expand the *Fourth Wall* notion to multiple actuators that can disrupt or intervene the user's headset camera experience, then the user must communicate as a performer with a possible audience not immersed in the XR world. However, the XR rig can play the immersive experience, but the real world is omnipresent due to the passing through component of the headset camera. Basically, to share a more social interaction, the visualization systems must enable good streaming video to projectors o much bigger screens.

In some ways the user adopts the role of an actor and can place eventually the animated holograms in the hands of the members of the audience or create a spot area that the audience can appreciate via projection. In essence the “Fourth Wall” metaphor acquires a more complex conceptual layer interlacing, allowing the audience to experience from outside the XR environment the tactile interaction with the holograms, but seen this particular action on one or a series of flat 4k screen.

3 IMMERSIVE NARRATIVE THROUGH INTELLIGENT PROGRAMMING

The immersive narrative is based on the content of the fresco's historical, situational, and symbolical backgrounds, representing how the story of the installation unfolds inside the immersive world. Basically, it is the content embodied in the layer of diegetic renderings deemed as a virtual wall, the one that establish the insights of communication as a visual network of cues. The users' engagement is driven by intelligent programmed components that articulates the rhythm and sequential flow of events through grabbing/throwing objects, using triggers, hand gestures, or the keys inputs of the devices' controller.

3.1 Navigational paths, cues, and visual insights

In the installation, several elements were developed to produce a dynamic navigation/exploration. When the user is approaching the virtual walls, scattered triggers deploy interactivity, including the physical banner's surfaces as well. Those triggers contain scripts of responses and interactions. However, the hands controllers for navigation and teleporting also contain tactile cues that make pop up 3d animated symbols to render the approachable content. The navigation to teleporting also involves the implementation of geographical situational spots where the layer of communication triggers or unfold options to produce awareness of an interactive object nearby. The navigation in the two modalities of immersion work on a dissimilar basis. In the VR modality the user is driven by teleporting and locomotion, backed by controllers or hand's interaction.



Figure 4. Navigational path and animated diegetic devices.

In the XR immersive mixed world, the user requires the physical walking to approach the places containing interactive objects. The headset can enable video through, and the path of navigation is consolidated in a physical surrounding. In this situation, any physical wall encompassing the installation becomes a layer for interaction as well, where the tactile action unravels objects containing levels of information. See Figure 5. In the XR world the intelligent programming is based on approachability, this means the closer to the physical object the camera rig, the better the response of activation from the overlapped 3D/holographic content installed on the spot. Another component in the VR world is remote triggering, using ray casts, and remote grabbing pressing triggers and grips from the hand's gestures action, or from the controller's keys.

3.2 Animated metaphorical diegetic devices

The animations structural messages become of critical importance once they are deployed within a conceptual level. They are capable not only in helping to enlighten effectively visual processes, but they are also indispensable to establish non-textual meanings, which are nevertheless appealing to intuitive and perceptual cognitive skills.

Perhaps a straightforward term would be of *Animated Metaphors*, to render properly an expanded narrative within a virtual world. The deconstruction of the whole “Wedding Chamber” through the immersive/expanded installation relies on animated metaphorical diegetic devices that recount the different stories from the original fresco. Those devices are, the medallions containing the roman emperor effigies, the *Temple of the Primeval Fortune*, *The Roca Pia*, Rome's imaginary depiction, and the *Myths of Orpheus, Ariadne, Hercules, and the Hydra* [5].

In the original fresco, Mantegna implements the construction of these precursory diegetic devices, but those remain motionless and formally embedded in the physical architectural structure of the frescos' room. Mantegna, in his virtuous compositional arrangements, anticipates the computing and cinematic use of these devices. In the installation, the goal was the extraction and articulation of these stories that belong to the whole fresco's rendition, into animated devices, that can be pulled out from the simulated layers of the VR world, or the expanded XR environment.

For a viewer that for the first-time visit the original “Wedding Chamber”, the reading of the whole story is a somewhat exhausting task, however with the assistance of the animated metaphorical diegetic devices, the narrative perhaps induced, it will unfold an efficient series of relations configuring the knowledge and intention that Andrea Mantegna wanted to trigger intuitively from the viewer, about the evident and concealed messages ingrained in his masterpiece

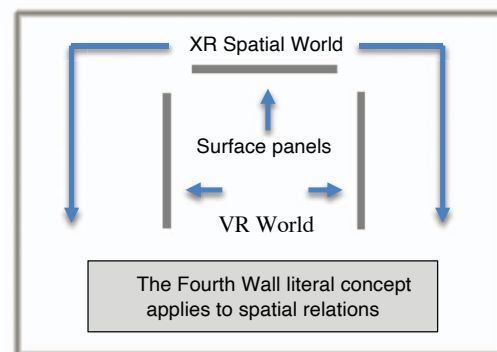


Figure 5. The VR/XR Immersive worlds with the metaphorical Fourth Wall concept containing 3D animated diegetic devices

3.3 The graphical sequence vs the frame rate action

In the installation, another component related with the moving imaging process that has been implemented is a “visual process through a graphical sequence”. Not perhaps deemed as a full frame rate animation, but it maintains a logical deployment of conceptual structure. The example of this technique is expressed in the deconstruction of the fresco’s walls panels to display color layouts that render how the chromatic application of pigments has been applied over the original fresco’s stucco. The renditions consist of using an interactive graphical sequence implementing an animation of flowing layering. In the immersive space, this configuration is very effective to grasp visually due to its spatial dimension. See Figure 6.

Layers can be triggered by a ray cast resulting in an animated cascade, but at the same time this cascading layering will also display animated concepts containing conventional frame by frame animation, turning the presence of graphical sequences, and frame rate animations, kinetic components that contribute to the diegetic narrative, and how this unfolds through a versatile and dynamic concept of moving image.



Figure 6. Layers animation in cascade graphical sequence

4 THE SUSPENSION OF JUDGEMENT WITHIN THE IMMERSIVE NARRATIVE USING DIEGETIC RENDERINGS

The intersection of computing and aesthetics is very relevant in the *Wedding Chamber* installation. The most important category as a byproduct of this crossing is the suspension of judgement which is the method of phenomenological reduction when an event is experienced. The installation main subject provides the building blocks for the implementation of an experimental phenomenological methodology of approaching each of the animated diegetic renderings of the installation. Each interactive object has animated layers that provide the users the instance of “suspended reflection” when the object is manipulated, then, each user can “feel” a very lively experience [6]. Each interactive object displays the animations with no text or documented references, to avoid a prejudgment, then after the experience is deployed or lived, the documentation related appears to produce in the user the insight of that knowledgeable moment afterwards. It is very important that the animated object also deploys spatial manipulation, then the users will be able to experience a body motion awareness in relation to the diegetic narrative of the animated object [7].

5 CONCLUSION

The methodological use of space-based animations in this project becomes of strategic relevance since the VR/XR technology set a consistent ground to experiment with innovative compositional principles within the realm of computing experiences. Extended Reality provides this fertile terrain to implement intuitively theoretical aesthetic categories that were deemed very unlikely to

fulfill on a dynamic and interactive basis before the advent of immersive emerging technologies. Additionally, this installation using animations as components to introduce levels of usability insights, can also configure not only an interdisciplinary usage, but it permits a channel of eloquent communication between art, technology, and science. The Diegetic Spatial Relations through animated objects, work in this project as an experimental but concrete example of how building blocks of implementation provided by emerging technologies serve to articulate cognitive domains that otherwise would only be theoretical. This installation is a work in progress, however the reflections, insights, and experiences that users provide, are very valuable data to refine and tweak the overall functionality and effectiveness of the project.

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